WINSTON MOH TANGONGHO

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EDUCATION

Northeastern University *Boston, MA*
Master of Science in Computer Science *May 2021*

Minnesota State University, Mankato *Mankato, MN*
Bachelor of Science in Computer Engineering *May 2018*

TECHNICAL KNOWLEDGE

- **Operating Systems**: Linux (Ubuntu), Windows OS, Mac OS
- **Languages**: C/C++, Java, Python, ASP.NET, Golang, C#, Flutter
- **Software**: Git, Visual Studio Code, Microsoft AZURE, Amazon AWS
- **Databases**: MongoDB, DynamoDB, SQL, COSMOS

WORK EXPERIENCE / ACADEMIC PROJECTS

Software Engineer 2, *Microsoft – Microsoft Advertising* *August 2021 – Present*

- Improved pipeline reliability and data freshness by building automated data freshness detection across 5+ daily production pipelines that self-recover from upstream delays without manual intervention, and implemented intelligent compute resource allocation using weighted heuristics that dynamically distributes resources across 30+ concurrent data processing jobs. This improved job throughput by reducing queue contention and eliminating stale data incidents that previously required on-call escalation.
- Strengthened data quality and production safeguards by building health monitoring for 10+ upstream data services, implemented deduplication and drift detection across billions of data count rows, and added publishing safeguards that block incomplete data drops. This prevented partial or corrupted data from reaching the production ad serving system that handles 100M+ requests daily and reducing data quality incidents to near-zero.
- Identified a critical scaling limitation in a high-throughput ad-ranking system nearing its configuration capacity (84% of hard limit), redesigned the underlying data architecture from fixed-size arrays to dynamic hash maps across multiple files in a latency-sensitive C++ codebase, validated zero performance regression across 5 production workloads, and expanded system capacity by 250x. This enabled continued business growth with no impact to serving latency.
- Designed and implemented a name-based ranker resolution mechanism for a real-time ML feature extraction pipeline in C++, enabling dynamic lookup of model ranker indices by name at runtime, reducing configuration fragility and decoupling classifier configs from hardcoded model internals across multiple evaluation and caching code paths, with comprehensive unit and E2E test coverage.
- Implemented brand safety and market consolidation reducing infrastructure by 33% by building a filtering system that removes 68% of non-compliant ad listings from brand-sensitive advertisers, and designed a bitmask-based market classification system that consolidated 3 separate regional index builds into a single pipeline — eliminating redundant compute, reducing end-to-end data latency, and enabling per-market filtering at query time across 2.2B+ indexed URLs.

Software Development Engineer Intern, *Amazon Web Services – AWS Support* *June 2020 – August 2020*

- Worked on a feature to integrate Amazon Support with a chat-based system to provide more flexibility for our premium customers to create and track their support cases. Utilized tools such as Java, CloudFormation and CloudWatch.
- Leveraged the AWS Step-Function together with a state-machine to periodically poll an AWS Service endpoint to get the status update of a support case. An AWS Lambda function was created to interact with the API endpoint.

Software Engineering Intern, *Datadog – Metrics Intake* *January 2020 – May 2020*

- Modified different parts of a consumer application (Golang production code) to add a Kubernetes readiness check on application startup. This improved traffic to the pods in case the application crashed during startup.
- Set-up autoscaling on a metric for two services which ran on Kubernetes with extra features to the existing HPA. This saved the company money, reduced the number of nodes by ~70 and made our systems more reactive to capacity changes.
- Created three new gRPC endpoints for an in-house application in Golang with batching for API calls to improve latency when communicating with the Postgres servers.
- Leveraged a Sarama client which implemented consumer groups by storing consumer offsets in Kafka instead of Zookeeper. This improved the consumer lag and removed a deprecated feature which stored consumer offsets in a different client.

Side Project – Apartment Tracker *February 2026 – Present*

- Built a cross-platform Flutter apartment management app with Finance tracking (daily income and expense management), building/user management, and dashboard analytics.
- Integrated Firebase messaging and local reminders for real-time notifications and alerts.
- Added camera/video dashboard support with web-safe fallback stubs for native media playback.